ASYMPTOTIC RS MCA

Abstract. We give a compact conditional proof of the asymptotic Reed–Solomon mutual-correlated-agreement frontier from the paid ledger developed in the v13 raw and Grande Finale manuscripts. The proof reduces the remaining primitive prefix problem to a fixed-density Boolean slice mapped by Vandermonde moments. After the image-normalized Fourier/Sidon cell is paid, any positive logarithmic moment excess gives a large high-energy Boolean fiber. Balog–Szemeredi–Gowers and quasicube difference growth forbid such a fiber, so primitive Q holds with only $\exp(o(n))$ loss. Combined with the stated add-back and lower-side inputs, every smooth row sequence covered by the conditional closed ledger has threshold $1-\rho-g^*(\rho,\log_2|\mathbb{B}|)+o(1)$. The closed-ledger package is stated explicitly, with the individual cell payments cited to the existing ledgers and with the C9 normalization exposed as an assumption.

1. Statement

Let $D \subseteq \mathbb{F}$ be an evaluation set of size n, and let

$$C = \mathrm{RS}_{\mathbb{F}}(D,k) = \{(f(t))_{t\in D} : \deg f < k\}.$$

For $a \in \{0,\dots,n\}$, a finite slope $\gamma \in \mathbb{F}$ is support-wise MCA-bad for a pair $r_1, r_2 \in \mathbb{F}^D$ at agreement a if some set $S \subseteq D$, $|S| \geq a$, and some $h \in \mathbb{F}[X]$, $\deg h < k$, satisfy $r_1 + \gamma r_2 = h$ on S, while r_1, r_2 are not simultaneously explained on the same S by two degree-$< k$ polynomials. Put

$$B_C^{\mathrm{MCA}}(a) = \max_{r_1,r_2} \#\{\gamma \in \mathbb{F} : \gamma \text{ is MCA-bad at agreement } a\}.$$

Then $\varepsilon_{\mathrm{mca}}(C, 1-a/n) = B_C^{\mathrm{MCA}}(a)/q_{\mathrm{line}}$, where q_{line} is the line-slope field used by the MCA sampler. This integer staircase convention is the one used in the v13 raw ledger [Cho26a] and in the Grande Finale reduction [Cho26b].

We consider smooth multiplicative or circle-type RS row sequences $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$ with $|D_n| = n$, $k_n/n \to \rho \in (0,1)$, and a base/generated alphabet $\mathbb{B}_n$ of bit-size $\beta_n = \log_2|\mathbb{B}_n| \to \beta$. For an agreement $a_n = k_n + 1 + w_n$, define the prefix average

$$\overline{N}_{n,a_n} = \binom{n}{a_n} |\mathbb{B}_n|^{-w_n}.$$

The entropy–subfield envelope is

$$g^*(\rho,\beta) = \sup\{g \in [0, 1-\rho] : H_2(\rho+g) \geq \beta g\},$$

where $H_2(x) = -x\log_2 x - (1-x)\log_2(1-x)$. The target threshold is $\delta_C^*(\varepsilon) = \sup\{\delta : \varepsilon_{\mathrm{mca}}(C,\delta) \leq \varepsilon\}$.

Definition 1.1 (Cells and first-match payment)**.** *A* cell *is a constructible subfamily of the MCA witness incidence space, or a bounded-complexity family of such subfamilies, stable under extension of the ground field. Cells are ordered. A slope is assigned to the first cell containing one of its witnesses. The cell is* paid *at agreement a_n if the number of slopes assigned to it is at most its printed budget, and a family is paid if the sum of its first-match budgets is $\exp(o(n))\overline{N}_{n,a_n}$.*

Lemma 1.2 (First-match disjointization)**.** *If ordered cells $\mathcal{C}_1,\dots,\mathcal{C}_J$ cover a set of bad slopes and the j-th first-match cell has budget U_j, then the covered slopes contribute at most $\sum_j U_j$ to $B_C^{\mathrm{MCA}}(a)$.*

Proof. For a fixed received line, assign each bad slope to the least-indexed cell containing a witness. These assigned slope classes are disjoint, and the j-th class is contained in the projection of $\mathcal{C}_j$ after earlier cells have been removed. Summing the budgets and then maximizing over received lines proves the claim. ☐

Definition 1.3 (Closed paid ledger)**.** *A smooth RS row sequence is* closed-ledger *at agreements* a_n *if the following first-match cells cover all MCA-bad witnesses except the primitive leaves of Theorem 2.1, and if their total contribution is* $\exp(o(n))\overline{N}_{n,a_n}$.

(C1) Quotient-pullback cells. *The support or locator descends along a nontrivial finite map* $D \to D'$*. These cells include quotient-remainder locators, divisor-block support ledgers, quotient-image lcm ledgers, and quotient pullback recursion.*
(C2) Chebyshev and dihedral cells. *On multiplicative or circle rows, the support is invariant under the dihedral action, or the locator factors through a Chebyshev or circle quotient such as* $z + z^{-1}$.
(C3) Planted-block cells. *A positive-density block is forced into the support or common zero set. The payment is the entropy cost of planting that block, together with the bounded quotient profile.*
(C4) Tangent and deep-center cells. *The received line lies in a tangent, secant, or high-contact stratum of the RS code variety, including the high-agreement tangent cell.*
(C5) Extension and descent cells. *The witness either descends to a smaller field or is controlled by a bounded extension-line or extension-pole parameter.*
(C6) Differential-locator and regular-rank cells. *The locator map has a rank drop, a ramification defect, or an overdetermined Hankel minor. These cells include regular closed-range Hankel packing, canonical rank-drop ledgers, and curve rank-drop ledgers.*
(C7) Saturation and effective-image-collapse cells. *The apparent support count is larger than the actual line-ray or codeword-ray image. The payment uses exact saturation identities and exact residual image ledgers.*
(C8) Balanced-core and split-pencil cells. *The residual witness lies in a balanced-core split-pencil chart. One-parameter primitive locator pencils are paid by the moving-root theorem; deficiency-one and identically split top charts are tangent-borne or eliminated.*
(C9) Fourier/Sidon cells. *In a primitive prefix leaf, an image-normalized heavy fiber with exponentially small additive closure is cut as a Fourier/Sidon cell. The moment is the* $L = |\operatorname{im}\Phi|$ *average from Theorem 2.1, not an ambient* $|K|^R$ *average: major arcs are routed to* (C1)–(C8)*, and minor arcs are paid by a Fourier/Sidon theorem stated at image scale.*

The remaining first-match complement is a disjoint union of primitive prefix leaves. Every such leaf has the Boolean fixed-density Vandermonde form of Theorem 2.1.

Theorem 1.4 (Conditional closed ledger package)**.** *For the smooth multiplicative and circle-type RS row sequences treated in the v13 raw ledger and in the Grande Finale notes, assume the cited* (C1)–(C8) *payments hold uniformly in the asymptotic frontier window, and assume the image-normalized C9 input of Theorem 3.2 for every surviving primitive leaf. Then the cells* (C1)–(C9) *form a closed paid ledger in that window.*

Proof. We record the existing ledger package rather than reprove each algebro-geometric count.

For (C1), the quotient-remainder lower and support ledgers, divisor-union support ledger, exact finite-parameter quotient-image ledger, affine/projective lcm ledgers, quotient pullback recursion, and bounded quotient census are proved in [Cho26a, quotient-remainder, quotient support, quotient-image, and bounded-census ledgers]. The corresponding first-match convention and quotient status cell are isolated in [Cho26b].

For (C2), exact Chebyshev fibers on circle twin-coset domains, circle quotient transport, and circle deployed ledgers are proved in [Cho26a, Chebyshev and circle sections]. Dihedral and Chebyshev cells are quotient cells with extra monodromy and are therefore paid by the same downstairs-support and image ledgers.

For (C3), planted quotient profiles, planted quotient-core lower counts, planted list-budget windows, and bounded active quotient order are established in [Cho26a, planted quotient profile and bounded quotient census]. These supply the entropy payment for positive-density forced blocks.

For (C4), the exact high-agreement tangent cell and tangent-borne split-kernel sections are proved in [Cho26a, high-agreement tangent exact cell and split-kernel tangent-borne theorem] and refined in [Cho26b, exact high-agreement tangent cell].

For (C5), the extension-pole deep-list and quotient-remainder floors, extension-line dimension-degree ledger, subfield confinement, and genuinely extension-valued line accounting are proved in [Cho26a, extension-pole and extension-line ledgers]; the full-extension cell target and extension-valued rank-one floor are recorded in [Cho26b].

For (C6), regular Hankel minors, canonical exact-agreement rank-drop ledgers, closed-ball rank-drop lcm ledgers, curve rank-drop ledgers, arbitrary-residual image ledgers, and the scanner residual ledger are proved in [Cho26a, Hankel, rank-drop, curve, residual-image, and scanner ledgers]. These are exactly the differential-locator and regular-rank payments.

For (C7), the exact saturation identity and line-ray saturation identity are proved in [Cho26b, saturation and line-ray saturation identities]; exact arbitrary-residual image and closed-ball residual image ledgers appear in [Cho26a, residual image ledgers]. Thus raw support overcounts are replaced by actual image counts.

For (C8), the interpolation-lattice split-pencil reduction, near-rational payment, deficiency-one split-test eliminant, split annihilator dictionary, tangent-borne split charts, boundary-profile identification with Q, base-field split-pencil floor, one-parameter moving-root theorem, and primitive one-pencil corollary are proved in [Cho26b, split-pencil and BC sections], with the balanced-core and split-pencil reductions also developed in [Cho26a, rank-one and split-pencil reductions].

For (C9), Fourier-flat leaves satisfy asymptotic Q and large-characteristic Fourier examples are proved in [Cho26b, Fourier-flat Q section]. The Sidon/Fourier operational cut, the split between Sidon-heavy and high-energy fibers, and the fact that major arcs route to algebraic cells while minor arcs are Fourier-flat are the moduli-ledger results of [Cho26d, Cho26c]. The C9 input consumed here is the image-normalized assumption in Theorem 3.2. Ambient max-fiber information may enter through Theorem 2.3, and ambient moment information may enter through Theorem 2.4; no unprinted bridge $A/L = \exp(o(N))$ is used. The image-normalized C9 input is used below in Theorems 3.1, 3.3 and 4.4.

Finally, first-match disjointization is Theorem 1.2 here and is also proved in the Grande Finale and moduli-ledger notes. The listed cells exhaust the nonprimitive branches of the v13 first-match atlas; after removing them, the remaining leaf is the primitive Boolean prefix slice. Summing the printed budgets gives $\exp(o(n))\overline{N}_{n,a_n}$, which is the closed paid ledger. ☐

Thus the row-specific geometry is packaged as an imported conditional ledger. The internal work below starts after those cell inputs, the image-normalized C9 assumption, the add-back profile input in Theorem 5.1, and the lower-side input in the proof of Theorem 1.5.

Theorem 1.5 (Asymptotic RS–MCA frontier)**.** *Let $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$ be a smooth RS row sequence as above. Assume the conditional closed paid ledger of Theorem 1.4 holds uniformly in every $o(n)$-window around the crossing $a_n = (\rho + g^*)n + o(n)$. Assume also the add-back profile input in Theorem 5.1 and a lower-side identity-prefix input with subexponential collision loss, or an equivalent collision-free identity-prefix floor. Then, for every target sequence with $\log_2(1/\varepsilon_n) = O(n)$ and fixed challenge normalization,*

$$\delta^*_{C_n}(\varepsilon_n) = 1 - \rho - g^*(\rho, \beta) + o(1),$$

with the evident replacement of β by $\lim \log_2 |\mathbb{B}_n|$ when the base alphabet varies.

Remark 1.6 (Current audit status)*.* The normalization repair in this version is limited to B1: C9 is consumed at image scale, and ambient/image moment transfers are printed explicitly. The note does not supply the missing C9 moduli/Fourier–Sidon source theorem, the B3 window-uniformity theorem, the add-back profile decomposition, or the lower-side pole-reservoir collision-loss proof.

Subject to the imported cell inputs, the image-normalized C9 assumption, the add-back profile input, and the lower-side input stated above, the rest of the paper proves Theorem 1.5. The proof is otherwise self-contained after the standard external additive-combinatorics facts stated in Section 4.

2. Primitive leaves

Definition 2.1 (Primitive prefix leaf). *A primitive leaf consists of a finite set T, $|T| = N$, an integer m with m/N bounded away from 0 and 1, a field K, nonzero weights $\rho(t) \in K^\times$, and columns*

$$v_t = \rho(t)(1, t, \dots, t^{R-1}) \in K^R, \qquad t \in T.$$

The active support family is a first-match residual slice

$$\Omega^\circ \subseteq \{x \in \{0,1\}^T : \sum_{t \in T} x_t = m\},$$

and the prefix map is $\Phi(x) = \sum_{t \in T} x_t v_t$. Write $\mathcal{S} = \operatorname{im} \Phi$, $L = |\mathcal{S}|$, $M = |\Omega^\circ|$, $\overline{N} = M/L$, and $F_s = \Omega^\circ \cap \Phi^{-1}(s)$. The leaf is at the frontier if $\log \overline{N} = o(N)$.

For $q = q_N \to \infty$, define

$$\Gamma_q^{\mathrm{ord}} = L^{-1} \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{\overline{N}} \right)^q.$$

Lemma 2.2 (Moment-max equivalence). *If $q \to \infty$ and $\log L = o(Nq)$, then*

$$L^{-1} \left(\max_s \frac{|F_s|}{\overline{N}} \right)^q \le \Gamma_q^{\mathrm{ord}} \le \left(\max_s \frac{|F_s|}{\overline{N}} \right)^q.$$

Hence $\Gamma_q^{\mathrm{ord}} \le \exp(o(Nq))$ is equivalent to $\max_s |F_s| \le \exp(o(N))\overline{N}$.

Proof. The upper bound replaces every summand by the maximum; the lower bound keeps a maximum summand. Taking q-th roots and using $\log L = o(Nq)$ gives the final assertion. □

Lemma 2.3 (Max-fiber ambient-to-image transfer). *Let $\Phi : \Omega^\circ \to G$ map into a finite abelian ambient group G of size A. Put $\mathcal{S} = \operatorname{im} \Phi$, $L = |\mathcal{S}|$, $M = |\Omega^\circ|$, and $F_s = \Omega^\circ \cap \Phi^{-1}(s)$. If*

$$\max_{y \in G} |\Omega^\circ \cap \Phi^{-1}(y)| \le \exp(o(N)) \frac{M}{A},$$

then

$$\max_{s \in \mathcal{S}} |F_s| \le \exp(o(N)) \frac{M}{L}.$$

Proof. Every image fiber is an ambient fiber, so

$$\max_{s \in \mathcal{S}} |F_s| \le \max_{y \in G} |\Omega^\circ \cap \Phi^{-1}(y)| \le \exp(o(N)) \frac{M}{A}.$$

Since $L \le A$, one has $M/A \le M/L$, proving the claimed image-normalized bound. □

Lemma 2.4 (Moment normalization identity). *Let $\Phi : \Omega^\circ \to G$ map into a finite abelian ambient group G of size A. Put $\mathcal{S} = \operatorname{im} \Phi$, $L = |\mathcal{S}|$, $M = |\Omega^\circ|$, $\overline{N} = M/L$, and $F_s = \Omega^\circ \cap \Phi^{-1}(s)$. For $q \ge 1$, define*

$$\Gamma_{\mathrm{img}}(q) = L^{-1} \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{M/L} \right)^q$$

and

$$\Gamma_{\mathrm{amb}}(q) = A^{-1} \sum_{y \in G} \left(\frac{|\Omega^\circ \cap \Phi^{-1}(y)|}{M/A} \right)^q.$$

Then

$$\Gamma_{\mathrm{amb}}(q) = \left(\frac{A}{L} \right)^{q-1} \Gamma_{\mathrm{img}}(q).$$

Consequently,

$$\Gamma_{\mathrm{img}}(q) = \left(\frac{L}{A} \right)^{q-1} \Gamma_{\mathrm{amb}}(q) \le \Gamma_{\mathrm{amb}}(q) \qquad (q \ge 1),$$

so an ambient upper bound is safe to use as an image upper bound. The reverse direction is unsafe without a printed bound on A/L, and the Fourier/Sidon payment consumed below is therefore stated directly at image scale.

Proof. The summands with $y \notin \mathcal{S}$ vanish. Since $M/A = (L/A)(M/L) = (L/A)\overline{N}$,

$$\begin{aligned}\Gamma_{\mathrm{amb}}(q) &= A^{-1} \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{M/A} \right)^q \\ &= A^{-1} \left(\frac{A}{L} \right)^q \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{\overline{N}} \right)^q \\ &= \left(\frac{A}{L} \right)^{q-1} L^{-1} \sum_{s \in \mathcal{S}} \left(\frac{|F_s|}{\overline{N}} \right)^q .\end{aligned}$$

This is the displayed identity. □

3. Sidon cut

For a finite F in an abelian group, let

$$E(F) = \#\{(a,b,c,d) \in F^4 : a - b = c - d\}, \qquad \Delta(F) = E(F)/|F|^3.$$

Thus $\Delta(F)$ is the probability that $a - b + c \in F$ for independent uniform $a, b, c \in F$.

For $\sigma > 0$, define the Sidon-heavy moment of a primitive leaf by

$$\Gamma^{\mathrm{sid}}_{q,\sigma} = L^{-1} \sum_{\Delta(F_s) \le e^{-\sigma N}} \left(\frac{|F_s|}{\overline{N}} \right)^q .$$

Definition 3.1 (Paid Fourier/Sidon cell)**.** *The Fourier/Sidon cell is paid if, for every fixed $\sigma > 0$ and every logarithmic q, one has $\Gamma^{\mathrm{sid}}_{q,\sigma} \le \exp(o(Nq))$. Equivalently, no positive-rate heavy fiber with exponentially small additive closure remains in the primitive residual. This is an image-normalized statement: $L = |\operatorname{im} \Phi|$, $\overline{N} = M/L$, and the average is over $\mathcal{S}$, not over an ambient group.*

Assumption 3.2 (Image-normalized C9 Fourier/Sidon input)**.** *Let $(T, m, K, \rho(t), v_t, \Omega^\circ, \Phi)$ be a primitive prefix leaf surviving cells* (C1)–(C8) *in the asymptotic frontier window. Write*

$$\mathcal{S} = \operatorname{im} \Phi, \qquad L = |\mathcal{S}|, \qquad M = |\Omega^\circ|, \qquad \overline{N} = M/L, \qquad F_s = \Omega^\circ \cap \Phi^{-1}(s).$$

The C9 moduli ledger input used by this note is the following image-scale payment: for every fixed $\sigma > 0$ and every logarithmic $q = q_N \to \infty$,

$$\Gamma^{\mathrm{sid}}_{q,\sigma} = L^{-1} \sum_{\Delta(F_s) \le \exp(-\sigma N)} \left(\frac{|F_s|}{\overline{N}} \right)^q \le \exp(o(Nq)).$$

Equivalently, the Fourier/Sidon cell is paid in the image normalization used by Theorem 2.1.

This is the C9 Fourier/Sidon input stated in the leaf normalization of Theorem 2.1. The present note does not prove the missing moduli ledger; it records the exact normalization consumed downstream. A proof of C9 may provide the displayed inequality directly, or may first prove a stronger image-scale minor-arc estimate after major arcs have been routed to (C1)–(C8). Ambient moment estimates can be consumed here only through Theorems 2.3 and 2.4.

In practice this cell is paid by Fourier flatness after major arcs have been routed to algebraic cells. The output needed here is an image-scale bound such as $L\mu(s) \le \exp(o(N))$ for all $s \in \mathcal{S}$, where $\mu = \Phi_* \operatorname{Unif}(\Omega^\circ)$, or directly the bound in Theorem 3.2. An ambient Fourier estimate does not imply this image-scale statement without one of the printed transfers above. In RS moment-curve leaves, large Fourier coefficients are algebraic major arcs: quotient, descent, dihedral, ramification, or effective-image-collapse phenomena. These are exactly the earlier paid cells in Theorem 1.3.

Proposition 3.3 (Energy extraction after the Sidon cut)**.** *Assume the Fourier/Sidon cell is paid. If $\Gamma^{\mathrm{ord}}_q \ge e^{\eta N q}$ for some fixed $\eta > 0$, then some level $s = s_N$ satisfies $|F_s| \ge e^{\eta N - o(N)}\overline{N}$ and $E(F_s) \ge |F_s|^3 e^{-o(N)}$.*

Proof. Fix $\sigma > 0$. Split Γ_q^{ord} into terms with $\Delta(F_s) \le e^{-\sigma N}$ and the complementary terms. The first part is $\Gamma_{q,\sigma}^{\mathrm{sid}} = \exp(o(Nq))$, so a complementary term has $(|F_s|/\overline{N})^q \ge e^{\eta Nq - o(Nq)}$. Hence $|F_s| \ge e^{\eta N - o(N)}\overline{N}$ and $\Delta(F_s) > e^{-\sigma N}$. Letting $\sigma \downarrow 0$ slowly along the sequence gives $E(F_s) \ge |F_s|^3 e^{-o(N)}$. □

4. Boolean additive combinatorics

We use the energy form of Balog–Szemeredi–Gowers [BS94, Gow98]. Its precise polynomial exponents are irrelevant here.

Theorem 4.1 (Balog–Szemeredi–Gowers)**.** *If A is a finite subset of an abelian group and $E(A) \ge |A|^3/K$, then some $A' \subseteq A$ satisfies $|A'| \ge K^{-C}|A|$ and $|A' - A'| \le K^C|A'|$, where C is absolute.*

We also use the quasicube sumset theorem of Matolcsi–Ruzsa–Shakan–Zhelezov [MRSZ20] in the sharpened form of Green–Matolcsi–Ruzsa–Shakan–Zhelezov [GMRSZ20]: if $U \subseteq \{0,1\}^d$, then $|P + Q + U| \ge |P|^{1/2}|Q|^{1/2}|U|$ for finite nonempty $P, Q \subseteq \mathbb{Z}^d$.

Theorem 4.2 (Quasicube difference growth)**.** *Every finite $A \subseteq \{0,1\}^N$ satisfies $|A-A| \ge |A|^{3/2}$.*

Proof. Apply the quasicube theorem with $U = A$, $P = -A$, and $Q = \{0\}$. □

Proposition 4.3 (No large high-energy Boolean fiber)**.** *There is no sequence $F_N \subseteq \{0,1\}^N$ with $|F_N| \ge e^{cN-o(N)}$ and $E(F_N) \ge |F_N|^3 e^{-o(N)}$ for fixed $c > 0$.*

Proof. By Theorem 4.1 with $K = e^{o(N)}$, a subset $F_N' \subseteq F_N$ has $|F_N'| \ge e^{cN-o(N)}$ and $|F_N' - F_N'| \le e^{o(N)}|F_N'|$. But Theorem 4.2 gives $|F_N' - F_N'| \ge |F_N'|^{3/2}$, so $|F_N'|^{1/2} \le e^{o(N)}$, a contradiction. □

Theorem 4.4 (Primitive Q after the Sidon cut)**.** *In every frontier primitive leaf with paid Fourier/Sidon cell, $\max_s |F_s| \le \exp(o(N))\overline{N}$.*

Proof. If not, Theorem 2.2 gives $\Gamma_q^{\mathrm{ord}} \ge e^{\eta Nq}$ for some fixed $\eta > 0$ along a subsequence and some logarithmic q. Theorem 3.3 then gives a fiber $F_s \subseteq \{0,1\}^T$ with $|F_s| \ge e^{\eta N - o(N)}\overline{N} = e^{cN-o(N)}$ and $E(F_s) \ge |F_s|^3 e^{-o(N)}$, because $\log \overline{N} = o(N)$. This contradicts Theorem 4.3. □

5. MCA compiler

Lemma 5.1 (Conditional first-match add-back)**.** *Assume the paid primitive leaves admit a first-match profile decomposition into $\exp(o(n))$ profiles, and that the normalized primitive leaf averages in those profiles sum to at most $\exp(o(n))\overline{N}_{n,a_n}$. Then, for a closed-ledger row, the full first-match prefix count after paid cells satisfies $\max_s N(s) \le \exp(o(n))\overline{N}_{n,a_n}$.*

Proof. Paid cells contribute $\exp(o(n))\overline{N}_{n,a_n}$ by closure. Primitive leaves satisfy Theorem 4.4. The stated profile-decomposition input bounds the sum of their normalized averages, and summing over the subexponential profile family preserves the same scale. The present note records this add-back dependency rather than proving the profile decomposition. □

Lemma 5.2 (Q pays shift pairs)**.** *If $\sum_s N(s) = M$, $\overline{N} = M/L$, and $\max_s N(s) \le \kappa\overline{N}$, then $M^{-1}\sum_s N(s)^2 \le \kappa\overline{N}$.*

Proof. $\sum_s N(s)^2 \le (\max_s N(s))\sum_s N(s) \le \kappa\overline{N}M$. □

Theorem 5.3 (Conditional closed-ledger upper bound)**.** *If a smooth RS row sequence is closed-ledger at a_n and satisfies the add-back input in Theorem 5.1, then*

$$B_{C_n}^{\mathrm{MCA}}(a_n) \le \exp(o(n))\binom{n}{a_n}|\mathbb{B}_n|^{-(a_n-k_n-1)}.$$

Proof. The nonprimitive cells are paid by definition. The primitive prefix leaves obey Q by Theorem 4.4 and add back by Theorem 5.1. The shift-pair second-moment ledger is paid by Theorem 5.2. First-match disjointization sums these disjoint slope classes, giving the displayed numerator bound. □

6. Lower side and frontier

We recall the identity-prefix pole construction in the form needed here. Let $m = k+1+w$, and for $S \subseteq D$, $|S| = m$, write $\ell_S(X) = \prod_{t\in S}(X-t)$. Pigeonholing the first w locator coefficients over $\mathbb{B}^w$ gives a prefix vector z with at least $\binom{n}{m}|\mathbb{B}|^{-w}$ supports. Choose a pole $\alpha \in \mathbb{F} \setminus D$, and set $U_z(X) = X^m + z_1 X^{m-1} + \cdots + z_w X^{m-w}$. The line

$$f_\alpha(x) = \frac{U_z(x)}{x-\alpha}, \qquad g_\alpha(x) = \frac{-1}{x-\alpha}, \qquad x \in D,$$

has the property that $f_\alpha + \zeta g_\alpha$ is code-explained on S exactly when $\ell_S(\alpha) = U_z(\alpha) - \zeta$. The word g_α is not explained by a degree-$< k$ polynomial on $k+1$ positions, so the corresponding slopes are MCA-bad. The lower side consumed by Theorem 1.5 assumes either a subexponential collision loss for evaluating $\ell_S(\alpha)$ in the pole-reservoir regime, or a replacement by a collision-free identity-prefix floor. Under that lower-side input, this gives

$$B_C^{\mathrm{MCA}}(m) \geq \exp(-o(n))\binom{n}{m}|\mathbb{B}|^{-w}.$$

This is the lower construction used in [Cho26a, Cho26b].

Proof of Theorem 1.5. Take $a_n = (\rho+g)n + o(n)$. Stirling gives

$$\log_2 \overline{N}_{n,a_n} = n(H_2(\rho+g) - \beta g) + o(n).$$

If $g > g^*(\rho,\beta)$ by a fixed amount, this exponent is negative by $\Omega(n)$, so Theorem 5.3 makes $B_{C_n}^{\mathrm{MCA}}(a_n)$ smaller than any target with the prescribed normalization after moving $o(n)$ agreements to absorb the $\exp(o(n))$ ledger overhead. Thus the safe radius reaches $1-\rho-g^*(\rho,\beta)-o(1)$. If $g < g^*(\rho,\beta)$ by a fixed amount, the identity-prefix construction gives exponentially many bad slopes, so the target fails at radius $1-\rho-g+o(1)$. Letting g approach g^* from both sides proves the claimed equality. □

Remark 6.1 (Finite adjacent rows). This is an asymptotic theorem. It intentionally absorbs $\exp(o(n))$ losses. The finite deployed adjacent pairs in [Cho26a, Cho26b] require the same first-match ledger with printed constants for every paid cell, Fourier/Sidon tail, and add-back step; those constants are not supplied by the asymptotic argument alone.

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